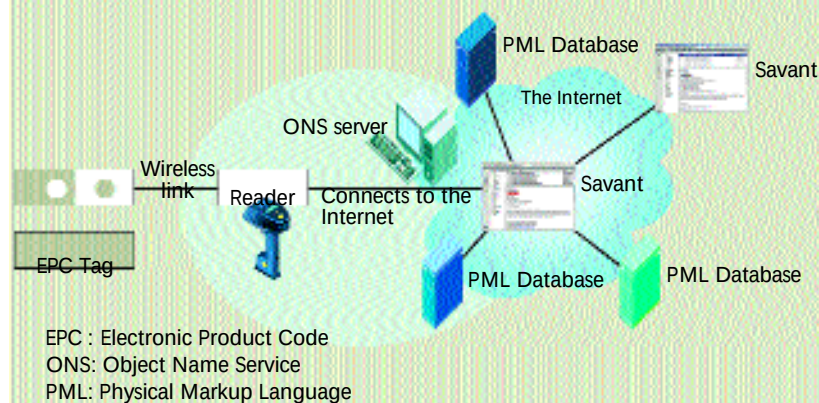


The Auto-ID Center's concept of the standardised RFID network



even considering putting RFID tags in currency notes to thwart counterfeiters.

In the future, your bank accounts will directly tie into your RFID-enabled cards, and the RFID-tagged goods you buy will be talking straight to the computers that make your bills. All you'd need to do in a supermarket is swipe your RFID-enabled card, pick up what you want and walk out. For some years now, it's been possible to pay highway toll charges in the US and Europe using RFID-based cards.

The problems

The biggest hurdle in RFID's path is cost. RFID readers start off at about Rs 23,000, and the cheapest tags come for about Rs 10. Barcodes are only about 50 paise. Also, there are problems of erroneous data capture. Initially, there were several incompatible RFID standards, which resulted in the technology not picking up in popularity. Other concerns include privacy. (See box, 'Privacy' on page 125)

With vast investments, there's been a deliberate move to standardise the technology and create cheap, workable solutions. A recent study by IDC, a global market intelligence and advisory firm in the IT and telecom industries, claims that by 2008, RFID should account for \$1.3 billion of the US retail supply chain.

That's a lot of money, but there's a world of information to be tapped in the retail supply chain. For this to work, though, the RFID system will have to be affordable, and that tags should be cheap enough to mass-produce. RFID experts have estimated that the magic number is the 5-cent per tag cost.

At MIT, researchers and industry sponsors formed the Auto-ID Center in 1999, which worked on creating a workable system to manage the data that's out there, for example, in the retail sector.

The result is the Electronic Product Code (EPC). EPC-compliant smart tags will hold just 96 bits of data; this includes a 40-bit serial number, and product and name information, amongst other parameters. Unlike conventional RFID tags, the Auto-ID framework emphasizes on minimal data—hence the 96-bit limit. The entire system will focus on storing the information on a commonly accessible computer network.

Think of it as the Internet of the RFID age, where the system will resolve the product information from the serial number using the Object Name Service (ONS), which is functionally similar to a DNS. The entire information is going to be coded in what's known as the Physical Markup Language, or PML.

The system works by querying the location of the servers hosting the information using the ONS, and then reading in the information on the PML server.

The entire system should be accessible with a computer connected to a reader, and an RFID tag-management software called Savant controls the RFID-related data read in.

The end result, as Auto-ID hopes, will standardise and bring down the cost of the RFID architecture, and eventually let EPC take over from the current UPC standard.

Patni's RFID pilot was based on the Auto-ID's framework, and it has a lab manned by supply-chain and RFID consultants. The lab lets them and their customers experiment with RFID deployment, and understand what RFID could mean for them.

RFID's grand entry

Technologies such as RFID will eventually move into the mainstream, perhaps through supply chains at first.

Experts opine that, for a while, RFID will co-exist with UPC. Once contentious issues are sorted out, there's every chance that we could see an example of cheap pervasive computing dominating our environment. Dhanuka says the RFID industry calls 2004 'the year of the pilots'. (2005 is expected to be the year that widespread RFID deployment begins.) So you won't need to wait too long—within the next couple of years, your neighbourhood store could well be tracking their supplies right from source. 

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